Lab 11

Configure LDAP as Secondary User Store

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **`<GW_HTTPS_PORT>` / `<GW_HTTP_PORT>`** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **`<APIM_HOME>`** – the folder where API Manager is extracted.

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue.

- **`<APACHEDS_HOME>`** – the folder where Apache Directory Server (ApacheDS) is extracted.

Training Objective

Learn how to add an LDAP directory as a secondary user store, so that developers can use their existing directory accounts to log in to WSO2 API Manager.

Business Scenario

PizzaShack wants to accelerate its digital transformation by providing an API Management platform to its developers.

High-Level Steps

- Start an LDAP server (Apache Directory Server)
- Add it to API Manager as a Read/Write LDAP secondary user store
- Add a user to the LDAP and log in with it

Before You Start

Identity Server 7.x no longer ships an embedded LDAP, so this lab uses Apache Directory Server (ApacheDS): a small Java LDAP server that runs on Windows, macOS and Linux with the JDK 21 you already installed. It starts with an admin account and with empty ou=users and ou=groups containers, which is all API Manager needs.

Shared training server: the trainer sets up the LDAP and the user store once (Parts 1 and 2). Each participant then does Part 3: adds their own user and logs in.

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Part 1 – Start the LDAP Server

1. Extract `apacheds-2.0.0.AM27.zip` (provided with the lab files). It extracts to a folder named `apacheds-2.0.0.AM28-SNAPSHOT`; that is expected. This folder is referred to as `<APACHEDS_HOME>`.

2. Start ApacheDS:

   macOS / Linux

   ```
   cd <APACHEDS_HOME>/bin
   sh apacheds.sh default start
   ```

   Example – macOS / Linux:

   ```
   cd ~/apacheds-2.0.0.AM28-SNAPSHOT/bin
   sh apacheds.sh default start
   ```

   To stop it later: `sh apacheds.sh default stop`

   Windows (Command Prompt) – keep the window open

   ```
   cd <APACHEDS_HOME>\bin
   apacheds.bat
   ```

   Example – Windows (Command Prompt) – keep the window open:

   ```
   cd C:\apacheds-2.0.0.AM28-SNAPSHOT\bin
   apacheds.bat
   ```

   The LDAP server listens on port **10389**. Its admin account is `uid=admin,ou=system` with the password `secret`.

Part 2 – Add the LDAP as a Secondary User Store

3. Start API Manager and log in to the Management Console as **admin**:

   ```
   https://<APIM_HOST>:<HTTPS_PORT>/carbon
   ```

   Example:

   ```
   https://localhost:9443/carbon
   ```

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4. Click **Main > User Stores > Add**. The **Add New User Store** page opens.

5. In the **User Store Manager Class** list, select **UniqueIDReadWriteLDAPUserStoreManager** (an LDAP that API Manager can read and write).

6. Enter the **Domain Name** (no underscore characters) and optionally a description. Domain name: `wso2.com` (API Manager shows it as **WSO2.COM**).

7. Change the following properties and leave all others at their default values. Expand the advanced section for **Immutable Attributes**.

Property	Value
Connection URL	`ldap://localhost:10389`
Connection Name	`uid=admin,ou=system`
Connection Password	`secret`
User Search Base	`ou=users,ou=system`
User Entry Object Class	`inetOrgPerson`
User ID Attribute	`entryUUID`
User ID Search Filter	`(&(objectClass=person)(entryUUID=?))`
Group Search Base	`ou=groups,ou=system`
Immutable Attributes (advanced)	`entryUUID`

   Why these values: ApacheDS stores users as inetOrgPerson entries under ou=users,ou=system and groups under ou=groups,ou=system. The default User ID Attribute (scimId) does not exist in ApacheDS, so the LDAP's own unique ID, entryUUID, is used instead.

8. Click **Add**. After about 15 seconds the **WSO2.COM** domain appears under **Main > User Stores > List**. If it does not, check the server log for a message about a missing property and correct it.

Part 3 – Add an LDAP User and Log In

9. Add a user to the new store: **Main > Users and Roles > Add > Add New User**, select the domain **WSO2.COM**, enter the user name `ldapuser1` and a password, and click **Next**. Select the roles **Internal/creator**, **Internal/publisher** and **Internal/subscriber**, and click **Finish**. The user is written to the LDAP. List the users of the WSO2.COM domain to check.



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10. Log in to the Publisher and the Developer Portal as `WSO2.COM/ldapuser1` (the domain prefix is part of the user name):

```
https://<APIM_HOST>:<HTTPS_PORT>/publisher
https://<APIM_HOST>:<HTTPS_PORT>/devportal
```

Example:

```
https://localhost:9443/publisher
https://localhost:9443/devportal
```

Expected Outcome

API Manager uses the LDAP as a secondary user store (domain WSO2.COM). Users created in that domain are stored in the LDAP and can log in to the Publisher and the Developer Portal.

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